Model Report for insurance_dataset:tabular

Generated on 07 Feb 2026, 02:36 • 1,338 original samples, 1,338 synthetic samples

Accuracy		
88.0% (94.1%)	Univariate	96.4% (98.0%)
	Bivariate	89.8% (94.3%)
	Trivariate	77.9% (84.5%)

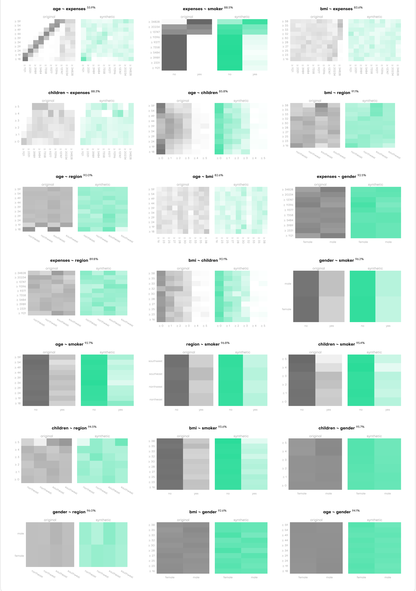
Similarity		
	Cosine Similarity	0.98051 (0.99811)
	Discriminator AUC	63.2% (50.0%)

Distances		
	Identical Matches	0.0% (0.0%)
	Average Distances	0.213
	DCR Share	51.3%
	NNDR Ratio	1.115 (1.000)

Correlations

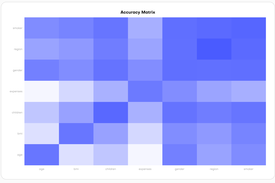
Univariate Distributions

Bivariate Distributions



Accuracy

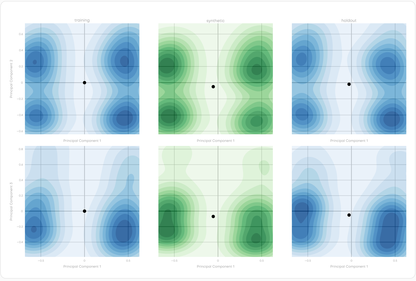
Column	Univariate	Bivariate	Trivariate
region	98.6%	93.0%	81.3%
smoker	97.2%	93.9%	84.2%
children	97.0%	91.6%	80.2%
gender	96.2%	94.5%	84.7%
bmi	95.4%	88.9%	76.1%
age	95.3%	83.5%	69.5%
expenses	94.8%	83.1%	69.1%
Total	**96.4%** (98.0%)	**89.8%** (94.3%)	**77.9%** (84.5%)



Explainer

Accuracy of synthetic data is assessed by comparing the distributions of the synthetic (shown in green) and the original data (shown in gray). For each distribution plot we sum up the deviations across all categories, to get the so-called total variation distance (TVD). The reported accuracy is then simply reported as 100% - TVD. These accuracies are calculated for all univariate, bivariate and trivariate distributions. A final accuracy score is then calculated as the average across all of these.

Similarity

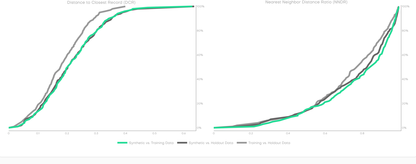


Explainer

These plots show the first 3 principal components of training samples, synthetic samples, and (if available) holdout samples within the embedding space. The black dots visualize the centroids of the respective samples. The similarity metric then measures the cosine similarity between these centroids. We expect the cosine similarity to be close to 1, indicating that the synthetic samples are as similar to the training samples as the holdout samples are.

Distances

	Synthetic vs. Training	Synthetic vs. Holdout	Training vs. Holdout
Identical Matches	0.0%	0.0%	0.0%
DCR Average	0.213	0.213	0.180
NNDR Min10	0.275	0.247	0.191
DCR Share	**51.3%** (of synthetic samples are closer to a training than to a holdout sample		
NNDR Ratio	**1.115** = (NNDR Min10 of Synthetic vs. Training) / (NNDR Min10 of Synthetic vs. Holdout)		



Explainer

Synthetic data shall be as close to the original training samples, as it is close to original holdout samples, which serve us as a reference. This can be asserted empirically by measuring distances between synthetic samples to their closest original samples, whereas training and holdout sets are sampled to be of equal size. A green line that is significantly left of the dark gray line implies that synthetic samples are closer to the training samples than to the holdout samples, indicating that the data has overfitted to the training data. A green line that overlays with the dark gray line validates that the trained model indeed represents the general rules, that can be found in training just as well as in holdout samples. The DCR share indicates the proportion of synthetic samples that are closer to a training sample than to a holdout sample, and ideally, this value should not significantly exceed 50%, as a higher value could indicate overfitting. The NNDR ratio is the ratio of the 10-th smallest NNDR for synthetic vs. training, divided by 10-th smallest NNDR for synthetic vs. holdout. Ideally, this value should be close to 1, indicating that the synthetic samples are in sparse as well as in dense regions just as close to the training samples as to the holdout samples.